

Arbitrage costs and the persistent non-zero CDS-bond basis: Evidence from intraday euro area sovereign debt markets*

Jacob Gyntelberg[†] Peter Hördahl[‡] Kristyna Ters[§] Jörg Urban[¶]

October 7, 2016

Abstract

We find evidence that in the market for euro area sovereign credit risk, arbitrageurs engage in basis trades between credit default swap (CDS) and bond markets only when the CDS-bond basis exceeds a certain threshold. This threshold effect is likely to reflect costs that arbitrageurs face when implementing trading strategies, including transaction costs and costs associated with committing balance sheet space for such trades. Using a threshold vector error correction model, we endogenously estimate these unknown trading costs for basis trades in the market for euro area sovereign debt. During the euro sovereign credit crisis, we find very high transaction costs of around 190 basis points, compared to around 80 basis points before the crisis. Moreover, we find that once the threshold has been exceeded, the basis reverts back towards its long-run equilibrium substantially faster than when the basis is below the threshold. Our findings help explain the persistent non-zero basis or spread between credit risk premia in CDS and sovereign bond markets and suggest that the significant increase in the basis during the crisis was due to sharply higher costs facing arbitrageurs in the market, as well as compensation for increased risk that the trade would go against them in the short run.

JEL classification: G12, G14 and G15.

Keywords: Sovereign credit risk, credit default swaps, price discovery, regime switch, intraday, arbitrage, transaction costs.

* The authors thank Darrell Duffie, Erik Theissen, and "OTC markets" participants at the SNB study center Gerzensee, participants at the University of Basel, the Bank for International Settlements, and the EFMA 2016 conference for useful discussions and comments. The views expressed in this paper are those of the authors and do not necessarily reflect those of the Bank for International Settlements. The research leading to these results has received funding from the Swiss National Science Foundation under career grant agreement no.: PMPDP1-158369.

[†] Danske Bank, Copenhagen, Denmark. email: jacob.gyntelberg@danskebank.dk

[‡] Bank for International Settlements, Hong Kong. email: peter.hoerdahl@bis.org

[§] University of Basel, Basel, Switzerland. email: kristyna.ters@unibas.ch

[¶] Bank for International Settlements, Basel, Switzerland. email: joerg.urban@bis.org

1 Introduction

The theoretical no-arbitrage condition between credit default swaps (CDS) and credit-risky bonds based on Duffie (1999) is a cornerstone for empirical research on price discovery in credit risk markets. This condition requires that CDS spreads and (par floating rate) spreads on bonds issued by the entity referenced in the CDS contract must be equal, as any discrepancy would present investors with an arbitrage opportunity. For this no-arbitrage condition to hold, markets must be perfect and frictionless. In practice, however, frictions and imperfections often make such arbitrage trades difficult and costly to varying degree. These imperfections include limited and time-varying liquidity across market segments, unavailability of instruments with identical maturity and payout structures, and the fact that some arbitrage trades require tying up large amounts of capital for extended periods of time. As a result, the difference between the CDS premium and the bond spread, the so-called basis, is typically not zero. Moreover, the basis can become sizeable and persistent in times of market stress.

A persistent non-zero CDS-bond basis is therefore likely to reflect the unwillingness of arbitrageurs to try to exploit it, unless the pricing mismatch is greater than the cost of undertaking the arbitrage trade. Empirically, we would therefore expect to see such arbitrage forces intensifying as the magnitude of the basis exceeds some level that reflects the costs that traders face in the market. This suggests that the adjustment process towards the long-run equilibrium is nonlinear, in that it differs depending on the level of the basis. In order to capture such behaviour, we extend the vector error correction model (VECM) which has been the convention in existing studies (see for example Blanco et al. (2005) and Zhu (2004) for corporates, Ammer and Cai (2007) for emerging markets, and Gyntelberg et al. (2013), Mayordomo et al. (2011), Palladini and Portes (2011), Fontana and Scheicher (2010) for euro area sovereigns) to a nonlinear set-up using a threshold VECM (TVECM).

The importance of frictions in credit risk modelling is well-known. However, only few empirical studies analyse the effect of frictions on the price discovery process for credit risk. Several papers conclude that for example liquidity affects corporate bond spreads significantly (e.g. Chen et al. (2007), Ericsson and Renault (2006) and Elton et al. (2001)). By contrast, other papers argue that CDS spreads reflect pure credit risk, i.e. that they are not significantly affected by liquidity (e.g. Longstaff et al. (2005)). However, there are numerous papers reporting that CDS spreads are too high to represent pure credit risk (e.g. Berndt et al. (2005), Blanco et al. (2005), Pan and Singleton (2005)). Tang and Yan (2007) find that the level of liquidity and liquidity risk are important factors in determining CDS spreads. Hull and White (2000) address the effects of market frictions from a theoretical point of view and determine conditions under which CDS prices are

affected. Longstaff et al. (2005) study price differences between CDS and bonds and attribute them to liquidity and counterparty risk. Also Zhu (2004) concludes that liquidity matters in CDS price discovery. Ammer and Cai (2007), Levy (2009) and Mayordomo et al. (2011) find evidence that liquidity (as measured by the bid-ask spread) is a key determinant for price discovery, but without explicitly modelling any market frictions. Tang and Yan (2007) focus on pricing effects in CDS and show that the liquidity effects on CDS premia are comparable to those on treasury and corporate bonds (Tang and Yan; 2007).

One of the key contributions of our paper to the existing literature on price discovery in credit markets is that, in contrast to all studies mentioned above, we allow for a nonlinear adjustment of prices in CDS and bond markets towards the long-run equilibrium. As mentioned, this will allow us to capture the possibility that arbitrageurs step into the market only when the trading opportunity is sufficiently profitable. Our TVECM approach can directly quantify the threshold beyond which such trading opportunities are seen by investors as 'sufficiently profitable'. Furthermore, our results show that even when markets in times of stress are liquid, the basis can widen as high market volatility makes arbitrage trades riskier, leading arbitrageurs to demand higher compensation (suggesting a higher threshold) before stepping into the market. This could explain why the basis became sizeable during the euro area sovereign debt crisis as it was subject to high volatility in a stressed market environment.

Our analysis relies on intraday price data for both CDS and bonds, allowing us to estimate the spread dynamics and the price discovery implications substantially more accurately than existing studies. Furthermore, the TVECM methodology allows us to quantify the region where arbitrageurs have little incentive to trade, as the costs associated with implementing arbitrage trades exceed the expected gain. We employ the TVECM methodology in an effort to capture the magnitude of the costs traders face, given that arbitrage costs such as funding and repo costs in the market for credit risk are opaque and not known a priori.

Our approach identifies thresholds in the magnitude of the CDS-bond basis, below which arbitrageurs are reluctant to step in. We also find that once the basis exceeds the threshold, the adjustment speeds towards the long run equilibrium intensify. This supports our assumption that arbitrageurs only step into the market when the trade becomes profitable. Our estimated average trading costs are around 80 basis points in the pre-crisis period. During the euro area sovereign debt crisis period we find increased arbitrage costs of around 190 basis points. The increased arbitrage costs during the crisis period can be explained with decreased liquidity in peripheral sovereign credit markets but also with a higher CDS-bond basis volatility. As arbitrageurs face the risk that the arbitrage trade will shift in the wrong direction, they will demand higher compensation

for undertaking the arbitrage trade in volatile markets. Thus, our findings help to explain the persistent non-zero basis in markets for sovereign credit risk as arbitrage costs prevent a complete adjustment.

The rest of the paper is structured as follows. Section 2 discusses in more detail the relationship between sovereign CDS and bonds. Section 3 explains our data, while Section 4 discusses the set-up and estimation of our TVECM. Section 5 provides the empirical results and Section 6 concludes.

2 Relation between sovereign CDS and bonds

2.1 Frictionless markets

In a frictionless market, the CDS premium should equal the spread on a par fixed-rate bond (issued by the same entity as referenced by the CDS) over the riskfree interest rate (Duffie (1999)). Both the CDS premium and the risky bond's yield spread is compensation to investors for being exposed to default risk, and must therefore be priced equally in the two market segments. However, for this to hold exactly, a number of specific conditions must be met, including that markets are perfect and frictionless, that bonds can be shorted without restrictions or cost, that there are no tax effects, etc. Any departures from this perfect environment will introduce potential wedges between the pricing of credit risk in CDS contracts and in bonds.

Moreover, given that floating rate notes are relatively uncommon, in particular for sovereigns, any comparison between CDS spreads and bond spreads based on fixed-rate bonds will introduce other distortions. Hence, the observed difference between the CDS premium and the bond spread, the basis, is typically not zero. To ensure proper comparability between the bond and the CDS, Gyntelberg et al. (2013) calculate synthetic asset swap spreads (ASW) for the bond leg of the basis, but nevertheless document that the basis persistently deviates from zero.

2.2 Markets with frictions

There are a number of recent papers that focus on the pricing of sovereign credit risk in the euro area, which all find that the theoretical no-arbitrage condition between CDS spreads and bond spreads does not hold (for example Gyntelberg et al. (2013), Fontana and Scheicher (2010), Arce et al. (2012), and Palladini and Portes (2011)). Gyntelberg et al. (2013) find that the basis across seven euro area sovereign entities¹ is almost always positive over the sample period for the 5 year and the 10 year tenor. Moreover, they

¹ France, Germany, Greece, Ireland, Italy, Portugal, Spain; 5- and 10-year tenor from October 2008 to end-May 2011

find that the basis varies substantially across countries, with means ranging from 74 to 122 basis points for the 5-year tenor, and from 58 to 175 basis points for the 10-year tenor. Empirical research on corporate credit risk also points towards a non-zero basis as shown for example in Nashikkar et al. (2011), Blanco et al. (2005) and Zhu (2004), and for emerging markets sovereign credit risk according to Ammer and Cai (2007).

The CDS market is a search market as the contracts are traded over-the-counter (OTC) where parties have to search for each other in order to bargain and match a trade. Therefore, market trading is not continuous in the sense that any amount can not be bought or sold immediately (Black; 1971). Moreover, other frictions and imperfections may make arbitrage trades difficult and costly. These imperfections include limited and time-varying liquidity in some or all market segments, unavailability of instruments with identical maturity and payout structures, and the fact that some arbitrage trades require tying up large amounts of capital for extended periods of time. As the costs associated with tying up space on banks' balance sheets have risen following the global financial crisis, this can represent a significant hurdle that traders face in the market. Furthermore, the no-arbitrage condition relies on the ability to short sell bonds, which is not always costless and sometimes even impossible due to illiquid markets. All of these imperfections contribute to explaining why the basis between CDS and bond spreads can deviate from zero, often substantially and persistently. However, we would expect to see arbitrage forces come into play if the basis becomes "too wide", thereby pushing it back towards zero.

Clearly, market liquidity conditions are crucial for investors that take positions in CDS and bond markets in order to exploit pricing differences, as high liquidity will tend to facilitate such transactions and keep the cost of doing so low. However, empirical evidence points towards the presence of liquidity frictions in CDS and bond markets which prevent a complete adjustment to the theoretical no-arbitrage condition. Arbitrageurs will only carry out a basis trade when the cost of the transaction is smaller than the expected gain from the trade. We would therefore expect to see stronger adjustment forces in CDS and bond markets when the basis exceeds some critical threshold. The size of the estimated threshold reflects the various arbitrage costs traders face in markets, including costs for illiquidity as well as for tying up costly capital of possibly long periods of time.

3 Data

For our empirical analysis we use intraday price quotes on CDS contracts and government bonds for France, Germany, Greece, Ireland, Italy, Portugal and Spain. We choose this group of countries because they include those that were most affected by the euro sovereign debt crisis. Germany is included as a near-riskfree reference country, and France which we consider as a low-risk control country. We use 5- and 10-year USD-denominated CDS

quotes for all countries in our sample. As documented in Gyntelberg et al. (2013), the 5-year segment is more liquid than the 10-year segment, particularly as the sovereign debt crisis intensified.

Our sovereign bond price data is provided by MTS (Mercato Telematico dei Titoli di Stato). The MTS data consists of both actual transaction prices and binding bid-offer quotes. The number of transactions of sovereign bonds on the MTS platform is however not sufficient to allow us to undertake any meaningful intraday analysis. Therefore, we will use the trading book from the respective domestic MTS markets.²

The CDS data consists of price quotes provided by CMA (Credit Market Analysis Ltd.) Datavision. CMA continuously gathers information on executable and indicative CDS prices directly from the largest and most active credit investors. After cleaning and checking the individual quotes, CMA applies a time and liquidity weighted aggregation so that each reported bid and offer price is based on the most recent and liquid quotes.³

We construct our intraday data on a 30-minute sampling frequency on our available data sets that spans from January 2008 to end-December 2011. The available number of indicative quotes for CDS does not allow higher data frequency than 30 minutes. The euro area sovereign CDS markets were very thin prior to 2008, which makes any type of intraday analysis before 2008 impossible (for a discussion please refer to Gyntelberg et al. (2013)).

When implementing our analysis we split the data into two sub-samples. The first sub-sample covers the period January 2008 to end-March 2010, and as such represents the period prior to the euro area sovereign debt crisis (van Rixtel and Gasperini; 2013). While this period includes the most severe phase of the financial crisis, including the default of Lehman Brothers, it is relatively unaffected by any major market concerns about the sustainability of public finances in euro area countries. The second sub-sample covers the euro area sovereign debt crisis period and runs from April 2010 to December 2011.

In order to accurately match the maturities and the cash flow structures of the CDS and the cash components for the measurement of the CDS-bond basis, we calculate intra-day asset swap (ASW) spreads based on estimated zero-coupon government bond prices according to Nelson and Siegel (1987). The use of ASW spreads is also in line with the practice used in commercial banks when trading the CDS-bond basis. By calculating ASW spreads we ensure that we are comparing like with like in our empirical analysis,

² We ignore quotes from the centralized European platform (market code: EBM), as quotes for government bonds on the centralised platform are duplicates of quotes on the domestic platforms. The MTS market is open from 8:15 to 17:30 local Milan time, preceded by a pre-market phase (7.30 to 8.00) and an offer-market phase (8:00 to 8:15). We use data from 8:30 to 17:30.

³ The CDS market, which is an OTC market, is open 24 hours a day. However, most of the activity in the CMA database is concentrated between around 7:00 and 17:00 London time. As we want to match the CDS data with the bond market data, we restrict our attention to the period from 8:30 to 17:30 local Milan time.

and we avoid introducing distortions by using imperfect cash spread measures, such as simple "constant maturity" yield differences.

An asset swap is a financial instrument that exchanges the cash flows from a given security - e.g. a particular government bond - for a floating market rate⁴. This floating rate is typically a reference rate such as Euribor for a given maturity plus a fixed spread, the ASW spread. This spread is determined such that the net value of the transaction is zero at inception. The ASW allows the investor to maintain the original credit exposure to the fixed rate bond without being exposed to interest rate risk. Hence, an asset swap on a credit risky bond is similar to a floating rate note with identical credit exposure, and the ASW is similar to the floating-rate spread that theoretically should be equivalent to a corresponding CDS spread on the same reference entity.

Finally, we note that using intraday data in our empirical analysis should enable us to obtain much sharper estimates and clearer results with respect to market mechanisms and price discovery compared to any analysis carried out with a lower data frequency (see Gyntelberg et al. (2013)).

Using the above methodology, we derive the intraday asset swap spreads for each country for the 5- and 10-year maturities (displayed in Appendix B). The corresponding CDS-bond basis series are shown in Figures 1 and 2.

4 Threshold vector error correction model (TVECM)

We begin our empirical analysis by examining the statistical properties of our spread time series. This analysis shows that the series are $I(1)$ and that the CDS and ASW series are cointegrated (see Appendix C and D). As a result, we can employ a vector error correction model (VECM) to study the joint price formation process in both markets. The VECM concept implies that any deviation from the long-run equilibrium of CDS and ASW spreads will give rise to dynamics that gradually will bring them back to the equilibrium due to an error correction mechanism. From the estimated error correction model one can then calculate measures that indicate which of the two markets is leading the price discovery process as well as examine the speed of adjustment towards the long-term equilibrium.

We extend the common VECM approach⁵ to a threshold vector error correction model (TVECM). Threshold cointegration was introduced by Balke and Fomby (1997) as a feasible mean to combine regime switches and cointegration. The TVECM model allows for nonlinear adjustments to the long-term equilibrium in CDS and bond markets. In our case, such nonlinear adjustment dynamics should be able to capture arbitrageurs' decisions to

⁴ See Appendix A. O'Kane (2000) and Gyntelberg et al. (2013) further discuss the mechanics and pricing of asset swaps.

⁵ As in e.g. Gyntelberg et al. (2013), and Fontana and Scheicher (2010), Blanco et al. (2005).

only step into the market when the basis exceeds some critical threshold, such that the expected profit exceeds the costs. As a result, adjustments to the long-term equilibrium would then be regime-dependent, with a relatively weak adjustment mechanism below the threshold (a 'neutral' regime) and a stronger adjustment mechanism above it.

4.1 Model specification

The TVECM approach allows the behaviour of y_t to depend on the state of the system. In our data, the basis for all reference entities is almost always positive. Hence, we expect to find at most two regimes with one threshold θ . One can formulate a two-regime TVECM as follows⁶:

$$\Delta y_t = \left[\lambda^L \beta^\top y_{t-1} + \Gamma^L(\ell) \Delta y_t \right] d_{Lt}(\beta, \theta) + \left[\lambda^U \beta^\top y_{t-1} + \Gamma^U(\ell) \Delta y_t \right] d_{Ut}(\beta, \theta) + \varepsilon_t \quad (1)$$

where the lower regime (specified by the index L) is defined as $\beta^\top y_{t-1} \leq \theta$, and the upper regime (specified by the index U) as $\beta^\top y_{t-1} > \theta$. Hence d_{Lt} and d_{Ut} are defined using the indicator functions $I(\cdot)$ as follows:

$$\begin{aligned} d_{Lt}(\beta, \theta) &= I(\beta^\top y_{t-1} \leq \theta) \\ d_{Ut}(\beta, \theta) &= I(\beta^\top y_{t-1} > \theta). \end{aligned}$$

$y_t = (CDS_t \quad ASW_t)^\top$ is a vector of price quotes for CDS and ASW at time t for a specific sovereign entity, while $\varepsilon_t = (\varepsilon_t^{CDS} \quad \varepsilon_t^{ASW})^\top$ is a vector of i.i.d. shocks. Equation (1) constitutes a vector autoregressive model in first-order difference with $\Gamma^j(\ell) = \sum_{k=1}^p \alpha^{j,k} \ell^k$ and ℓ as lag operator, p as number of VAR lags, $j \in \{L, U\}$, and an additional error correction term $\beta^\top y_{t-1}$. This error correction term represents the long-term equilibrium of the two time series which has to be stationary by construction (Johansen; 1988). The number of VAR lags is determined using the Schwarz information criterion. The error correction term can in general be written as $\beta^\top y_{t-1} = (CDS_{t-1} - \beta_0 - \beta_1 ASW_{t-1})$. We constrain β_1 to 1 which is motivated by our no-arbitrage discussion in Section 2. β_0 represents then the persistent non-zero basis. The estimates are disclosed in Appendix E while our constrained error correction term is $I(0)$.

The speed of adjustment parameters characterize to what extent the price changes in $\Delta y_t = (\Delta CDS_t \quad \Delta ASW_t)^\top$ react to deviations from the long-term equilibrium. In case price discovery takes place only in the bond market we would find a negative and statistically significant λ_1^j and a statistically insignificant λ_2^j , as the CDS market would

⁶ for a derivation of the TVECM see for example Balke and Fomby (1997)

adjust to correct the pricing differentials from the long-term relationship. In other words, in this case the bond market would move ahead of the CDS market as relevant information reaches investors. Conversely, if λ_1^j is not statistically significant but λ_2^j is positive and statistically significant, the price discovery process takes place in the CDS market only - that is, the CDS market moves ahead of the bond market. In cases where both λ 's are significant, with λ_1^j negative and λ_2^j positive, price discovery takes place in both markets.

We expect to find the speed of adjustment parameters to indicate that arbitrageurs are engaging in CDS-ASW basis trades if the basis exceeds the threshold θ . In a market with a positive basis (CDS > ASW), arbitrageurs bet on a declining basis and will short credit risk in the bond market and go long credit risk in the CDS market, i.e. sell the bond *and* sell the CDS (Gyntelberg et al.; 2013).⁷ The predominantly positive basis throughout our sample suggests the presence of at most one threshold.

Moreover, we expect to find higher thresholds (trading costs) in times of market stress. This can be explained by the fact that when the basis is subject to increased volatility, the risk that any arbitrage trade moves in the wrong direction in the short or medium run increases. Therefore, arbitrageurs will demand higher compensation for taking such positions in times when the basis volatility is high, and as a result we expect to find higher thresholds.

4.2 Estimating the threshold

As discussed in Section 4, the positive basis in our sample suggests the presence of at most one threshold. In order to test for the presence of a threshold effect, we follow the method proposed by Hansen and Seo (2002) who extend the literature by examining the case of an unknown cointegrating vector.⁸ They implement maximum likelihood estimation (MLE) of a bivariate TVECM with two regimes. Their algorithm involves a joint grid search over the threshold and the cointegrating vector while using the error-correction term as the threshold variable (see Equation (1)). All coefficients are allowed to switch between these two regimes. Only the cointegrating vector β remains fixed across all regimes, by construction.

As in Hansen and Seo (2002) we estimate the model while imposing the following additional constraint:

$$\pi_0 \leq P(\beta^\top y_{t-1} \leq \theta) \leq 1 - \pi_0 \quad (2)$$

⁷ In case of a negative basis (ASW > CDS), arbitrageurs bet on an increasing basis while carrying out the reverse trade. In markets where the basis regularly would fluctuate between being positive and negative, we would expect to find a 3-regime TVECM. With a lower regime $\beta^\top y_{t-1} < \theta^1$, a middle regime (neutral regime) $\theta^1 \leq \beta^\top y_{t-1} \leq \theta^2$, and an upper regime $\theta^2 < \beta^\top y_{t-1}$.

⁸ Balke and Fomby (1997) and Tsay (1989) transform the TVECM specification into a univariate regression while the cointegrating vector is known a priori.

where $\pi_0 > 0$ is a trimming parameter and P is the share of observations in each regime. This constraint allows us to identify a threshold effect only if the share of observations in each regime is greater than π_0 . If this condition is not met, the model reduces to a linear VECM.

Andrews (1993) argues that setting π_0 between 0.05 and 0.15 are typically good choices. As we use intraday data of the order of 10,000 observations, we set the trim to $\pi_0 = 0.10$, which will still ensure an adequate number of observations in both regimes.

The maximum likelihood estimators (MLE) $(\hat{\lambda}^L, \hat{\lambda}^U, \hat{\Sigma}, \hat{\beta}, \hat{\theta})$ are the values which maximize $\mathcal{L}_n(\lambda^L, \lambda^U, \Sigma, \beta, \theta)$. For computational reasons, Hansen and Seo (2002) suggest to hold (β, θ) fixed and compute the constrained MLE for $(\lambda^L, \lambda^U, \Sigma)$, which corresponds in their model set-up to a grid search over a 2-dimensional space (β, θ) . We also have to search over a 2-dimensional space (β_0, θ) , where β_0 is our intercept, which is not present in the Hansen and Seo (2002) model setup.

4.3 Statistical testing for a threshold

Once a threshold has been identified, the next step is to determine whether the estimated threshold θ is statistically significant. Under the null hypothesis there is no threshold, so the model reduces to a conventional linear VECM where $\lambda^L = \lambda^U$. The two regime TVECM is the alternative hypothesis $\mathcal{H}_1$ with $\lambda^L \neq \lambda^U$ under the constraint in Equation (2). The linear VECM under $\mathcal{H}_0$ is nested in Equation (1), hence, a regular LM test with an asymptotic $\chi^2(N)$ -distribution can be calculated based on Equation (1). However, the LM test can only be applied if the cointegrating vector β and the threshold variable θ are known a priori (Hansen and Seo; 2002). While the point estimate of β under $\mathcal{H}_0$ is $\hat{\beta}$ from the linear model, there is no estimate of θ under $\mathcal{H}_0$. This implies that there is no distribution theory for the parameter estimates and no conventionally defined LM statistic.

We follow Hansen and Seo (2002) and perform two different bootstrap methodologies in order to estimate the asymptotic distribution for our model specification in Equation (1). First, we implement a non-parametric bootstrap on the residuals, called the "fixed regressor bootstrap", which resamples (Monte-Carlo) the residuals from the estimated linear VECM. The second bootstrap methodology, called the "residual bootstrap", is parametric.

We consider our model as threshold cointegrated if we can reject the null hypothesis of a linear VECM by either the "residual bootstrap" or the "fixed regressor bootstrap" methodology. Our results are robust with respect to the choice of the trimming parameter.

4.4 Measure of price discovery

The VECM-based measures of price discovery defined by Hasbrouck (1995) can be straightforwardly applied to a TVECM setup. From Equation (1) we calculate the independent set of values HAS_1 and HAS_2 (Hasbrouck; 1995) for each regime, whereby we define HAS as the average of HAS_1 and HAS_2 .

The Hasbrouck measure is by construction confined to the closed interval $[0,1]$. This makes an interpretation straightforward, namely $HAS > 0.5$ can be interpreted as the CDS market contributing more to price discovery than the cash market. Similarly, $HAS < 0.5$ means that the bond (ASW) market contributes more to price discovery.

Finally, we are interested in examining the speed of adjustment towards the long-term equilibrium in each regime. As the CDS and ASW spreads in the bivariate VECM share a common stochastic trend, the speed of adjustments of the cointegrating residual to the long-run equilibrium can be used to determine the impulse response function (Zivot and Wang; 2006). The vector error correction mechanism directly links the speed of adjustment of CDS and ASW spreads to the regime dependent cointegrating error u_t^j which follows an implied AR(1) process:

$$\begin{aligned} u_t^j &= (1 + \lambda_1^j - \beta_1 \lambda_2^j) u_{t-1}^j + \varepsilon_t^{CDS} - \beta_1 \varepsilon_t^{ASW} \\ &= (1 + \lambda_1^j - \lambda_2^j) u_{t-1}^j + \varepsilon_t^{CDS} - \varepsilon_t^{ASW} \equiv \phi^j u_{t-1}^j + \varepsilon_t^{CDS} - \varepsilon_t^{ASW}, \end{aligned} \quad (3)$$

where we have set β_1 to 1 in the second line of the equation. The half-life of a shock for each regime, hl^j , can now be calculated from the AR(1) coefficient ϕ^j as:

$$hl^j = \frac{\ln(0.5)}{\ln(\phi^j)}. \quad (4)$$

5 Results

In this section we first present results for the period before the euro area sovereign debt crisis (January 2008 to end-March 2010). These are followed by our findings using data for the sovereign debt crisis period (April 2010 to December 2011).

As a general result, we find a functioning relationship between the CDS market and the bond market during both samples. In cases where we find threshold cointegration, the adjustment process towards the long-term equilibrium is faster in the upper regime compared to the lower regime, in line with our reasoning on the behaviour of arbitrageurs. The estimated thresholds in the pre-debt-crisis period are around 80 basis points. For the second sub-period (sovereign debt crisis) we find much higher thresholds of approximately 190 basis points. The two to three times higher thresholds during the crisis period are

in line with our expectations, as markets were subject to stress and liquidity dried out in peripheral sovereign credit markets. Arbitrageurs therefore demand higher compensation for undertaking an arbitrage trade as the risk of the trade moving in the wrong direction is elevated. The estimated thresholds are shown in Figures 1 and 2.

5.1 Results for the pre-debt-crisis period

The results for the first sub-sample from January 2008 to end-March 2010, i.e. prior to the euro area sovereign debt crisis, confirm our assumption that arbitrage trading intensifies in CDS and bond markets once some basis threshold is exceeded. In the lower (neutral) regime we find as expected either no adjustment dynamics, or speed of adjustments that are much smaller in magnitude than in the upper regime. The price discovery results for the 5-year and 10-year tenor are presented in Table 1.

For the 5-year tenor, we fail to find threshold effects for most countries. As expected we find more thresholds for the less liquid 10-year tenor in the pre-crisis period, because less liquid market segments have more frictions and higher arbitrage costs and are thus more likely to exhibit multi-regime behaviour.

Our results are supportive of our hypothesis regarding arbitrageurs behaviour in markets with frictions. We find either faster adjustment dynamics towards the long-term equilibrium in the upper regime compared to the lower regime, or no adjustments in the lower regime, in which case the dynamics in the lower regime is a simple VAR. Further, for most reference entities in the 10-year tenor adjustment dynamics in the lower regime exhibit the wrong sign (see Table 1). The half-lives of any basis widening are also either significantly shorter in the upper regime compared to the lower regime (the only exception is France). This suggests that arbitrage trading activity is much higher in the upper regime and therefore pricing differences due to credit risk shocks are reabsorbed much faster once the threshold is exceeded (Table 2). Typically, the upper regime can be viewed as an extreme regime as the bulk of observations is in most cases concentrated in the lower (neutral) regime. This is due to the fact that if the basis moves into the upper regime, arbitrageurs will start trading which will move the basis rapidly back into the lower regime.

Figure 1: CDS-ASW basis, 5 year tenor

The basis is the difference between the CDS spread and the ASW spread expressed in basis points for the period from January 2008 until December 2011. The figure shows data with 30-minute sampling frequency. Due to the Greek debt restructuring the data for Greece ends in September 2011. During the crisis period a simple VECM for Germany (superscript ⁺) is a better model fit than any threshold model based on maximum likelihood estimation.

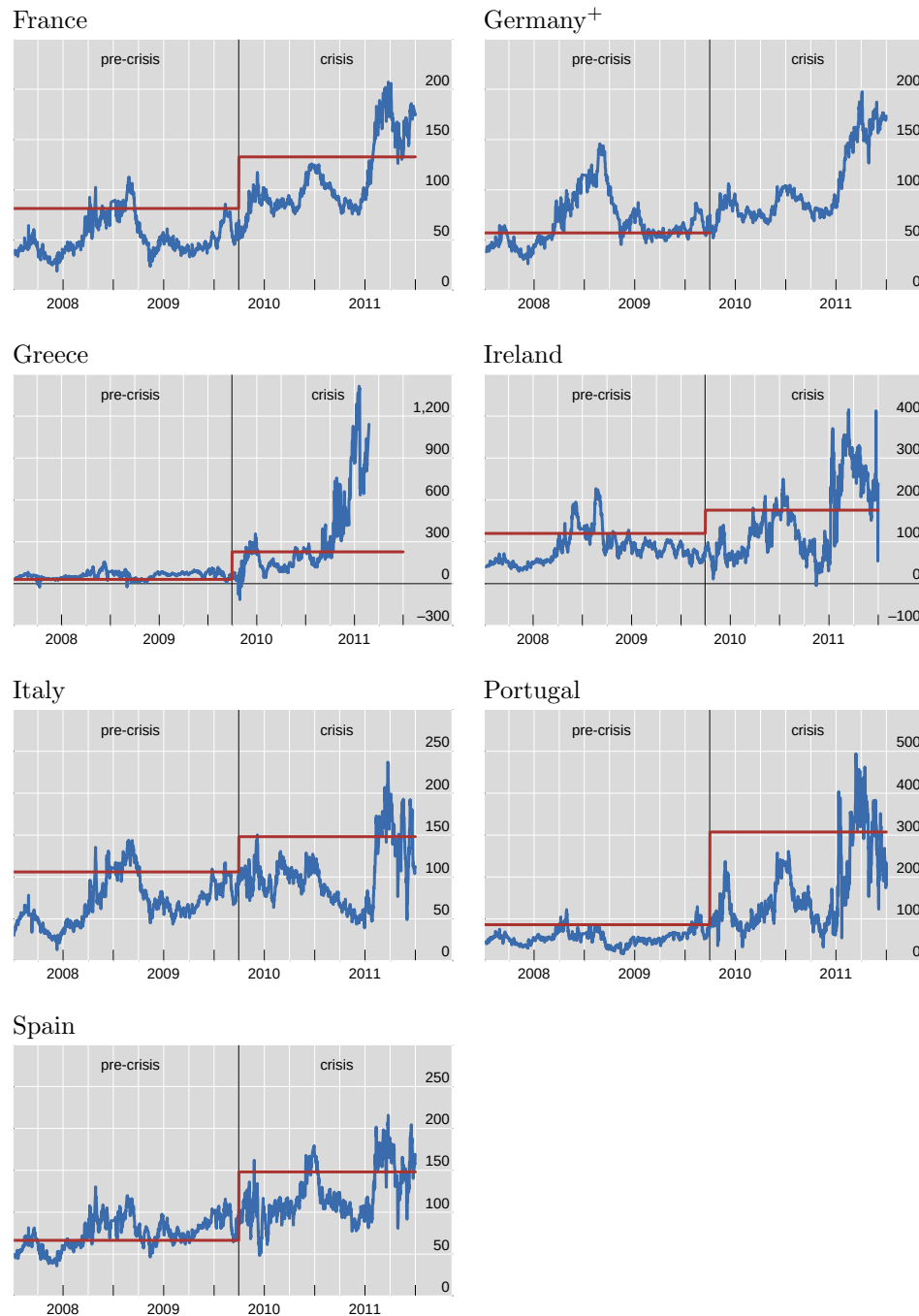
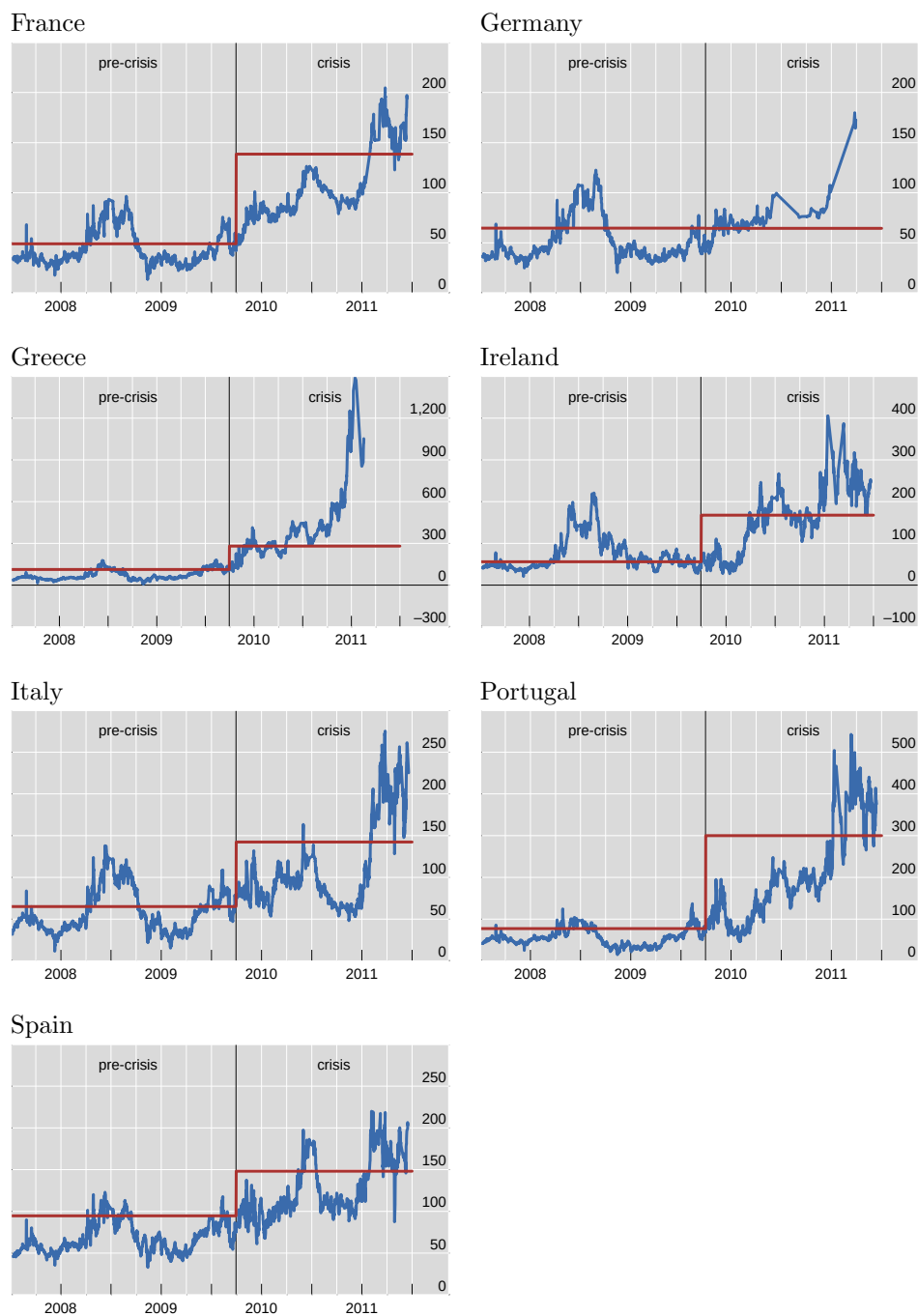


Figure 2: CDS-ASW basis, 10 year tenor

The basis is the difference between the CDS spread and the ASW spread expressed in basis points for the period from January 2008 until December 2011. The figure shows data with 30-minute sampling frequency. Due to the Greek debt restructuring the data for Greece ends in September 2011.



In most cases, we find that the CDS market leads the price discovery process for sovereign credit risk in the upper regime of the 10-year tenor as displayed in Table 1 with HAS ratios that are significantly above 0.5 in most cases. Spain is an exception as the bond market leads the pricing of credit risk in the upper regime.

Table 1: Price discovery TVECM - pre-crisis period

This table reports the price discovery analysis for the period from January 2008 to end-March 2010. The values of the VECM coefficients λ are expressed in units of 10^{-4} . HAS is defined as the average of HAS₁ and HAS₂ (Hasbrouck; 1995). The threshold θ is presented in basis points. The superscripts U and D denote the upper and lower regime, respectively. The average of the thresholds in the last line of each table takes only the significant thresholds into account.

Panel A - 5-year tenor

Sovereign	θ	HAS ^U	λ_1^U	λ_2^U	HAS ^L	λ_1^L	λ_2^L	obs.
France	81.4	0.63	-14.93*	-21.60**	0.17	-1.81*	1.40	87.2%
Germany	57.0**	0.81	0.19	0.60	0.93	2.22	-7.46*	16.9%
Greece	30.1	0.94	-10.69	105.04***	0.67	-67.32*	254.66	12.6%
Ireland	120.1*	0.71	5.23	6.24	0.82	2.28	-5.66	87.6%
Italy	106.1**	0.06	-7.13	-1.48	0.01	6.29	1.01	83.17%
Portugal	86.0*	0.07	-54.32*	-1.36	0.73	13.25	30.95**	89.9 %
Spain	66.4*	0.24	-25.96*	13.77	0.19	-14.90	7.46	18.3%
average	87.1							

Panel B - 10-year tenor

Sovereign	θ	HAS ^U	λ_1^U	λ_2^U	HAS ^L	λ_1^L	λ_2^L	obs.
France	49.1	0.90	13.45	79.29**	0.00	-7.22**	0.00	66.2%
Germany	64.7	0.92	0.54	3.87	0.55	1.92*	-2.01**	78.5%
Greece	113.0	0.72	-16.40	23.58**	0.05	20.97***	5.70	81.7%
Ireland	56.0*	0.93	-2.23	4.93**	0.66	4.18	-6.70	39.1%
Italy	65.1**	0.84	-2.92	6.79*	0.21	6.07*	-4.55	55.5%
Portugal	77.2**	0.75	-15.33	24.14**	0.06	14.39**	-4.15	81.1%
Spain	94.7**	0.01	-23.98**	-2.05	0.51	14.66**	7.74	90.0%
average	73.3							

Table 2: Half-life of shocks in days - pre-crisis period

This table reports the half-life of shocks of 5-year and 10-year CDS and ASW for the period from January 2008 to end-March 2010. The half-lives of shocks are expressed in days, and are calculated using the impulse response function to a one unit shock on the cointegrating error, using Equations (3) and (4). In case of no significance of both λ_i coefficients in a regime we have a VAR model. In case the speed of adjustments have a wrong sign we do not report the VECM based half-life of shocks.

	5-year tenor		10-year tenor	
Sovereign	lower	upper	lower	upper
France	212.7	-	53.3	-
Germany	-	VAR	-	VAR
Greece	5.7	3.6	-	16.3
Ireland	VAR	VAR	VAR	78.1
Italy	VAR	VAR	-	56.7
Portugal	12.4	7.1	-	15.9
Spain	VAR	14.8	-	16.0

5.2 Results for the euro area sovereign debt crisis period

The results for the euro area sovereign debt crisis period that spans from April 2010 to end-December 2011 show that arbitrage forces continue to functioning despite the turbulent market conditions. Arbitrageurs step into the market once the basis exceeds the trading costs (thresholds). During the crisis period we find either no, or much slower adjustment speeds in the lower regime, where significant thresholds are identified (Table 3). These results are in line with our findings for the pre-crisis period. We find thresholds that are two to three times higher than in the pre-crisis period with an average of 190 basis points.

The sharply higher estimated trading costs can be explained by decreased liquidity in peripheral sovereign credit markets, in combination with a markedly higher volatility of the basis (see Appendix F). As arbitrageurs face the risk that the arbitrage trade will go against them in the short- to medium-run, they will demand a higher compensation for undertaking the trade in volatile markets. The pre-crisis period is characterised by much lower basis volatilities across the countries in our sample.

For the crisis period we cannot make a general conclusion on which market typically leads in the price discovery for credit risk as we find mixed results. For the 5-year tenor, we find CDS leadership for Portugal and Greece (upper regime). Results for the French and Irish cases suggest bond leadership. For the 10-year tenor we find CDS leadership in

the upper regime for France and Greece, whereas bonds dominate for Germany. In the lower regime we find either bond leadership or no error correction at all.

All half-lives are displayed in Table 4. The abbreviation VAR in the table indicates that the estimates suggest no adjustment towards the long term equilibrium. The few cases where the speed of adjustments have a wrong sign (either CDS or ASW move away from the long-term equilibrium) the half-lives are not reported as the implied dynamics are unstable.

Table 3: Price discovery TVECM - crisis period

This table reports the price discovery analysis for intraday data on a 30-minute sampling frequency from the TVECM for the period from April 2010 to end-December 2011 for the 5- and 10-year tenor. In the case of Germany, 5 year tenor (superscript ⁺), the VECM is a better fit compared to any threshold model based on maximum likelihood estimation. For further details see Table 1.

Panel A - 5-year tenor

Sovereign	θ	HAS ^U	λ_1^U	λ_2^U	HAS ^L	λ_1^L	λ_2^L	obs.
France	132.8**	0.01	-65.84***	-8.51	0.16	-1.58	1.30	77.1%
Germany ⁺	-	-	-	-	-	-	-	-
Greece	227.7**	0.55	123.78	498.80*	0.27	-10.45*	11.56	89.5%
Ireland	175.5*	0.14	-53.32***	33.74	0.01	-10.84***	-0.31	70.6%
Italy	148.2***	0.02	-15.16	-0.85	0.76	-6.95	16.78	87.5%
Portugal	307.3***	0.78	-10.45	75.54**	0.03	-16.52*	3.94	87.9%
Spain	148.27	0.12	-17.14	2.48	0.75	-31.68	70.51***	80.7%
average	198.3							

Panel B - 10-year tenor

Sovereign	θ	HAS ^U	λ_1^U	λ_2^U	HAS ^L	λ_1^L	λ_2^L	obs.
France	138.6*	0.99	4.44	25.98***	0.02	-18.46**	-3.43	86.0%
Germany	64.5*	0.13	-13.12**	5.16	0.06	37.13*	-9.58	36.9%
Greece	280.0***	0.57	10.00	15.16*	0.93	-1.64	4.42	44.6%
Ireland	167.7	0.26	-12.66	4.26	0.00	19.02	0.31	62.2%
Italy	142.3*	0.13	-22.94	-4.92	0.91	9.50	17.30	89.0 %
Portugal	300.1*	0.88	-8.43	-19.72	0.83	4.24	7.99	89.9%
Spain	95.4	0.17	-13.95	-5.29	0.18	-293.89	76.58	16.1%
average	185.1							

Table 4: Half-life of shocks in days - crisis period

This table reports the half-life of shocks of 5-year and 10-year CDS and ASW for the period from April 2010 to end-December 2011. For further details see Table 2. In the case of Germany, 5 year tenor, the VECM is a better fit compared to any threshold model based on maximum likelihood estimation, hence no half life is reported.

	5-year tenor		10-year tenor	
Sovereign	lower	upper	lower	upper
France	VAR	5.8	20.8	14.8
Germany			-	29.3
Greece	36.8	0.8	VAR	25.4
Ireland	35.5	7.2	VAR	VAR
Italy	VAR	VAR	VAR	VAR
Portugal	23.3	5.1	VAR	VAR
Spain	5.4	VAR	1.29	VAR

5.3 Basis trade gain

The basis trade gain (BTG_{adj}) represents the daily risk and cost adjusted potential basis trade gain, expressed in basis points, that an arbitrageur can typically expect in the upper regime, as implied by the model estimates. In the upper regime, the arbitrageur will bet on a declining basis while going short credit risk in the bond market and going long credit risk in the CDS, i.e. by selling the bond and selling the CDS (Gyntelberg et al.; 2013). The overall costs the arbitrageur faces in the market is our estimated threshold θ from Tables 1 and 3. We assume that the typical basis widening in the upper regime is the mean value of the basis in the upper regime. When deducting the overall costs θ from the typical basis widening, we get the expected basis trade gain denoted as $E(\text{basis trade gain})$ in Equation (5).

In the short run, the arbitrageur faces the risk of the trade moving in the wrong direction which is directly proportional to the basis volatility. To generate a risk-adjusted daily basis trade gain ratio BTG_{adj} , we adjust the daily potential trading gain by the daily basis volatility (vola_d). Daily volatilities of the basis are displayed in Appendix F.

$$\text{BTG}_{\text{adj}} = \frac{E(\text{basis trade gain})}{\text{hl}_d^j} \cdot \frac{1}{\text{vola}_d} \quad (5)$$

The subscript d represents the normalisation on a daily frequency.

As expected, we find higher risk-adjusted trading gain ratios in the upper regime, which are presented in Table 5. Further, they remain more or less stable between the pre-crisis and the crisis period. This is in line with our expectations, as our BTG_{adj} is a risk adjusted measure. As before, in the few cases where the speed of adjustments have a wrong sign we do not report the BTG_{adj} .

Table 5: Daily risk and cost adjusted basis trade gain

This table reports the daily risk and cost adjusted basis trade gain (BTG_{adj}) from Equation (5) on a typical basis widening trade (arithmetic mean of the basis in the upper regime), in basis points. The few cases where the speed of adjustments have wrong signs are left empty. Further, no adjustments towards the long term equilibrium is a VAR thus the BTG_{adj} is zero.

Panel A - 5-year tenor

	pre-crisis		crisis	
Sovereign	BTG_{adj}	E(basis trade gain)	BTG_{adj}	E(basis trade gain)
France		12.53	4.64	34.63
Germany	0.00	24.70		
Greece	2.06	35.58	29.84	231.37
Ireland	0.00	42.22	3.22	82.59
Italy	0.00	15.39	0.00	26.27
Portugal	0.72	12.53	2.38	60.97
Spain	0.75	18.56	0.00	21.30

Panel B - 10-year tenor

	pre-crisis		crisis	
Sovereign	BTG_{adj}	E(basis trade gain)	BTG_{adj}	E(basis trade gain)
France		19.12	1.24	24.30
Germany	0.00	24.70	0.44	11.84
Greece	0.45	23.76	2.16	143.94
Ireland	0.20	41.89	0.00	55.95
Italy	0.20	24.44	0.00	60.13
Portugal	0.36	13.64	0.00	94.09
Spain	0.35	8.40	0.00	33.87

6 Conclusion

The persistence of a positive basis between sovereign CDS and sovereign bond spreads in the euro area points to the presence of arbitrage costs that prevent a complete adjustment of market prices to the theoretical no-arbitrage condition of a zero basis. These include

transaction costs and costs associated with committing balance sheet space for implementing arbitrage trades. Using a TVECM modelling approach, we are able to quantify these unobservable costs and study their properties.

We find that the adjustment process towards the long-run equilibrium intensifies once the CDS-bond basis exceeds a certain level/threshold. Above this estimated threshold, the trade becomes profitable for arbitrageurs while below the threshold, arbitrageurs have no incentive for trading as the costs they face are higher than the expected gain from the trade. As a result, we find much faster adjustment dynamics towards the long-term equilibrium once the estimated threshold is exceeded (upper regime) compared to the lower regime, and the half-life of any basis widening is therefore significantly shorter in the upper regime compared to the lower regime. This supports our assumption that arbitrageurs step in and carry out basis trades only when the expected gain from the arbitrage trade is greater than the trading costs.

During the euro sovereign credit crisis in 2010-11, we find very high estimated transaction cost of around 190 basis points, compared to around 80 basis points before the crisis. This increase was likely due to higher costs facing arbitrageurs in the market, as well as higher risk that the trade would go against them due to substantially more volatile market conditions. In response, arbitrageurs demanded higher compensation for undertaking such trades during the crisis, resulting in higher thresholds. In line with this, we find that risk-adjusted trading gain ratios (expected one-day profit from implementing an arbitrage trade, corrected for trading costs, divided by daily basis volatility) remained more or less stable between the pre-crisis and the crisis period, suggesting that market participants adjusted their behaviour as market conditions changed to equilibrate risk-adjusted expected returns.

References

- Ammer, J. and Cai, F. (2007). Sovereign cds and bond pricing dynamics in emerging markets: Does the cheapest-to-deliver option matter?, *International Finance Discussion Papers 912*, Board of Governors of the Federal Reserve System.
- Andrews, D. W. K. (1993). Tests for parameter instability and structural change with unknown change point, *Econometrica* **61**: 821–1414.
- Arce, O., Mayordomo, S. and Pena, J. I. (2012). Credit-risk valuation in the sovereign cds and bonds markets: Evidence from the euro area crisis, *Documentos de Trabajo CNMV* **53**: 1–37.
- Balke, N. S. and Fomby, T. B. (1997). Threshold cointegration, *International Economic Review* **38**(3): 627–645.

- Berndt, A., Douglas, R., Duffie, D., Ferguson, M. and Schranz, D. (2005). Measuring default risk premia from default swap rates and edfs, *Working paper*, Cornell University.
- Black, F. (1971). Toward a fully automated stock exchange, *Financial Analysts Journal* **27**: 28–35.
- Blanco, R., Brennan, S. and Marsh, I. W. (2005). An empirical analysis of the dynamic relation between investment-grade bonds and credit default swaps, *The Journal of Finance* **LX**(5): 2255–2281.
- Chen, L., Lesmond, D. A. and Wei, J. (2007). Corporate yield spreads and bond liquidity, *Journal of Finance* **62**: 119–149.
- Duffie, D. (1999). Credit swap valuation, *Financial Analysts Journal* **55**: 73–87.
- Elton, E. J., Gruber, M. J., Agrawal, D. and Mann, C. (2001). Explaining the rate spread on corporate bonds, *Journal of Finance* **56**: 247–277.
- Ericsson, J. and Renault, O. (2006). Liquidity and credit risk, *Journal of Finance* **61**: 2219–2250.
- Fontana, A. and Scheicher, M. (2010). An analysis of euro area sovereign cds and their relation with government bonds, *EZB Working Paper Series 1271*, European Central Bank.
- Gyntelberg, J., Hoerdahl, P., Ters, K. and Urban, J. (2013). Intraday dynamics of euro area sovereign cds and bonds, *BIS Working Papers 423*, Bank for International Settlements.
- Hansen, B. E. and Seo, B. (2002). Testing for two-regime threshold cointegration in vector error-correction models, *Journal of Econometrics* **110**(2): 293–318.
- Hasbrouck, J. (1995). One security, many markets: Determining the contributions to price discovery, *Journal of Finance* **50**(4): 1175–1199.
- Hull, J. and White, A. (2000). Valuing credit default swaps i: No counterparty default risk, *Journal of Derivatives* **8**(1): 29–40.
- Johansen, S. (1988). Statistical analysis of cointegrated vectors, *Journal of Economic Dynamics and Control* **12**: 231–254.
- Levy, A. (2009). The cds bond basis spread in emerging markets: Liquidity and counterparty risk effects (draft).

- Longstaff, F., Mithal, S. and Neis, E. (2005). Corporate yield spreads: Default risk or liquidity? new evidence from credit-default swap market, *Journal of Finance* **60**: 2213–2253.
- Mayordomo, S., Pena, J. I. and Romo, J. (2011). The effect of liquidity on the price discovery process in credit derivatives markets in times of financial distress, *European Journal of Finance* **17**: 851–881.
- Nashikkar, A., Subrahmanyam, M. G. and Mahanti, S. (2011). Liquidity and arbitrage in the market for credit risk, *Journal of Financial and Quantitative Analysis* **46**: 627–656.
- Nelson, C. R. and Siegel, A. F. (1987). Parsimonious modeling of yield curves, *Journal of Business* **60**(4): 473–489.
- O’Kane, D. (2000). Introduction to asset swap spreads, *Technical report*, Lehman Brothers.
- Palladini, G. and Portes, R. (2011). Sovereign cds and bond pricing dynamics in the euro-area, *NBER Working Paper Series 17586*, The National Bureau of Economic Research.
- Pan, J. and Singleton, K. J. (2005). Default and recovery implicit in the term structure of sovereign cds spreads, *Working paper*, MIT and Stanford.
- Phillips, P. C. B. and Ouliaris, S. (1990). Asymptotic properties of residual based tests for cointegration, *Econometrica* **58**(1): 165–193.
- Tang, D. Y. and Yan, H. (2007). Liquidity and credit default swap spreads, *Research paper*, Kennesaw State University and University of South Carolina.
- Tsay, R. S. (1989). Testing an modeling threshold autoregressive processes, *Journal of the American Statistical Association* **84**(405): 231–240.
- van Rixtel, A. and Gasperini, G. (2013). Financial crises and bank funding: recent experience in the euro area, *BIS Working Paper 406*, Bank for International Settlements.
- Zhu, H. (2004). An empirical comparison of credit spreads between the bond market and the credit default swap market, *BIS Working Paper 160*, Bank for International Settlements.
- Zivot, E. and Wang, J. (2006). *Modeling Financial Time Series with S-PLUS*, Springer-Verlag New York, Inc.

A Asset Swap Spreads

Specifically, the ASW is the fixed value A required for the following equation to hold⁹ (O’Kane (2000))

$$\underbrace{100 - P}_{\text{Upfront payment for bond asset in return for par}} + C \underbrace{\sum_{i=1}^{N_{\text{fixed}}} d(t_i)}_{\text{Fixed payments}} = \underbrace{\sum_{i=1}^{N_{\text{float}}} (L_i + A) d(t_i)}_{\text{Floating payments}}, \quad (6)$$

Interest rate swap

where P is the full (dirty) price of the bond, C is the bond coupon, L_i is the floating reference rate (e.g. Euribor) at time t_i , and $d(t_i)$ is the discount factor applicable to the corresponding cash flow at time t_i .

In order to compute the ASW A several observations and simplifications have to be made. First, in practice it is almost impossible to find bonds outstanding with maturities that exactly match those of the CDS contracts and second, the cash-flows of the bonds and the CDS will not coincide. To overcome these issues, in what follows we use synthetic asset swap spreads based on estimated intraday zero-coupon sovereign bond prices. Specifically, for each interval and each country, we estimate a zero-coupon curve based on all available bond price quotes during that time interval using the Nelson-Siegel (1997) method. With this procedure we are able to price synthetic bonds with maturities that exactly match those of the CDS contracts, and we can use these bond prices to back out the corresponding ASW. As this results in zero coupon bond prices, we can set C in Equation (6) to zero.

A CDS contract with a maturity of m years for country j at time interval k of day t , denoted as $S_j(t_k, m)$, has a corresponding ASW $A_j(t_k, m)$:

$$100 - P_j(t_k, m) = \sum_{i=1}^{N_m} (L_i(t_k) + A_j(t_k, m)) \cdot d(t_k, t_i), \quad (7)$$

where $P_j(t_k, m)$ is our synthetic zero coupon bond price.

For the reference rate L_i in Equation (7), we use the 3-month Euribor forward curve to match as accurately as possible the quarterly cash flows of sovereign CDS contracts. We construct the forward curve using forward rate agreements (FRAs) and Euro interest rate swaps. We collect the FRA and swap data from Bloomberg, which provides daily (end-of-day) data. 3-month FRAs are available with quarterly settlement dates up to 21 months ahead, i.e. up to 21×24 . From two years onwards, we bootstrap zero-coupon swap rates from swap interest rates available on Bloomberg and back out the corresponding implied

⁹ This assumes that there is no accrued coupon payment due at the time of the trade; otherwise, an adjustment factor would need to be added to the floating payment component.

forward rates. Because the swaps have annual maturities, we use a cubic spline to generate the full implied forward curve, thereby enabling us to obtain the quarterly forward rates needed in Equation (7).

Given our interest in intraday dynamics, we follow Gyntelberg et al. (2013) and generate estimated intraday Euribor forward rates by assuming that the intraday movements of the Euribor forward curve are proportional to the intraday movements of the German government forward curve.¹⁰ To be precise, for each day, we calculate the difference between our Euribor forward curve and the forward curve implied by the end-of-day Nelson-Siegel curve for Germany.¹¹ We then keep this difference across the entire curve fixed throughout that same day and add it to the estimated intraday forward curves for Germany earlier on that day to generate the approximate intraday Euribor forward curves. This approach makes the, in our view, reasonable assumption that the intraday variability in Euribor forward rates will largely mirror movements in corresponding German forward rates.

Finally, we need to specify the discount rates $d(t_k, t_i)$ in Equation (7). The market has increasingly moved to essentially risk-free discounting using the overnight index swap (OIS) curve. We therefore take $d(t_k, t_i)$ to be the euro OIS discount curve, which is constructed in a way similar to the Euribor forward curve. For OIS contracts with maturities longer than one year, we bootstrap out zero-coupon OIS rates from interest rates on long-term OIS contracts. Thereafter, we construct the entire OIS curve using a cubic spline. We use the same technique as described above to generate approximate intraday OIS discount curves based on the intraday movements of the German government curve.

¹⁰ Euribor rates are daily fixing rates, so we are actually approximating the intraday movements of the interbank interest rates for which Euribor serves as a daily benchmark.

¹¹ Here we use the second to last 30-minute interval, because the last trading interval is occasionally overly volatile.

B CDS and ASW spreads

Figure B.1: CDS and ASW spreads in basis points

The figures are based on data with a 30-minute sampling frequency.

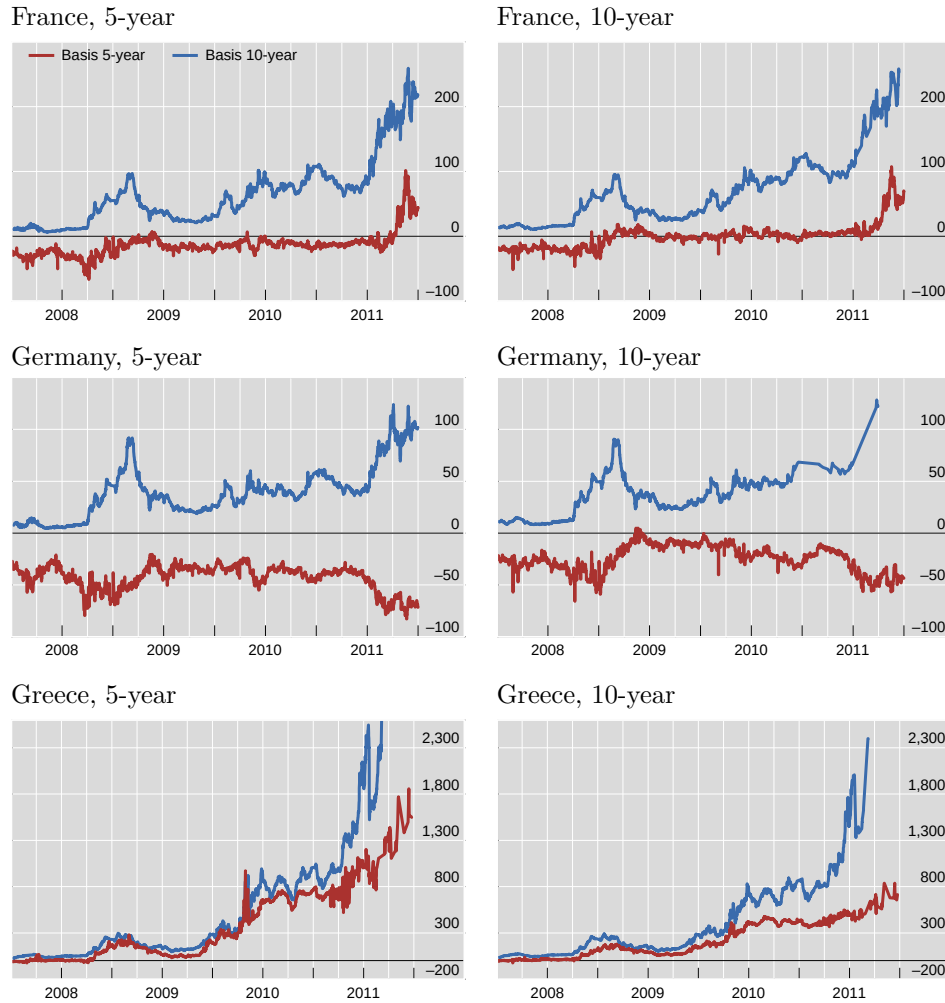
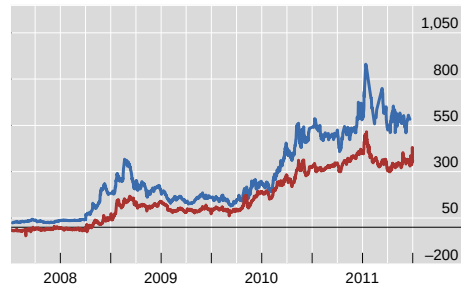


Figure B.1: (Cont.) CDS and asset swap spreads

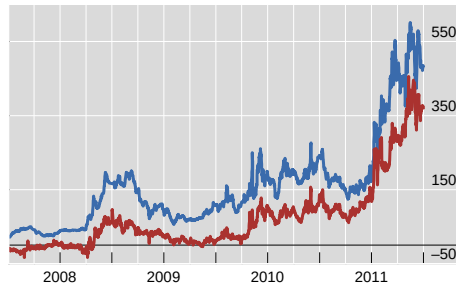
Ireland, 5-year



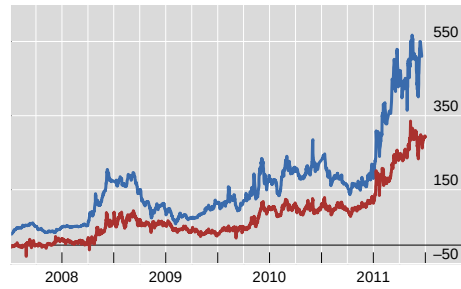
Ireland, 10-year



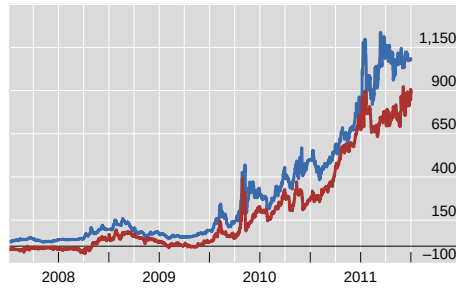
Italy, 5-year



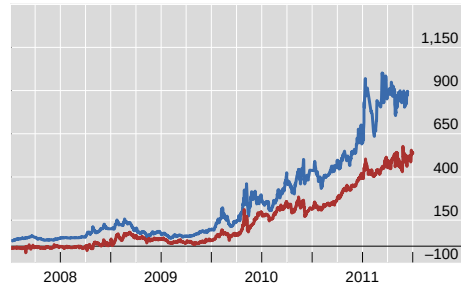
Italy, 10-year



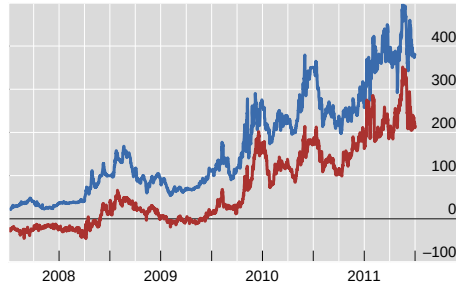
Portugal, 5-year



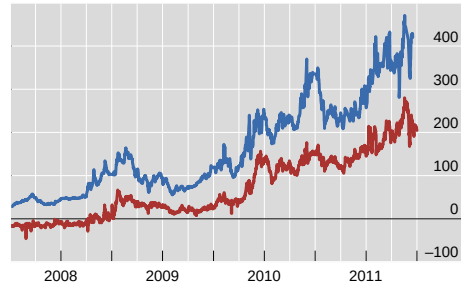
Portugal, 10-year



Spain, 5-year



Spain, 10-year



C Unit root and stationarity tests

We test for unit roots and stationarity in the CDS and ASW time-series using the following three methods:

1. the Augmented Dickey-Fuller (ADF) test,
2. the Phillips-Perron (PP) test and
3. the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test.

The null hypothesis of the ADF and PP test states: *the series has a unit root*. The null hypothesis of the KPSS test is: *the series is stationary*. Therefore, if our CDS and ASW data are $I(1)$ time series, we should be unable to reject the null hypothesis in levels for the ADF and PP test and reject H_0 under the KPSS test, and vice versa for first differences.

Based on these three different tests we conclude that both the CDS and the asset swap spreads have a unit root for both tenors and periods (pre-crisis and crisis).

Our findings in Tables C.1 and C.2 show that for none of the CDS series in levels we are able to reject the null hypothesis of a unit root using either the ADF or the PP test. For the asset swap spread series the null is rejected for a few countries and tenors in levels using both the ADF and PP test. The KPSS rejects stationarity for all countries and both maturities. Test results for the first differenced spread data show that for all test methods we reject the unit root hypothesis across the board, indicating that all series are integrated of order one. To conserve space, we do not show these test results, but they are available from the authors on request.

Table C.1: Unit root and stationarity tests in levels - pre-crisis

The table reports the statistics of unit root and stationarity tests for the period from January 2008 to end-March 2010. The ADF and PP test for a unit root under the null hypothesis. For the KPSS test, the null is stationarity, and the 0.01, 0.05 and 0.10 critical values for the test statistics are 0.739, 0.463 and 0.347, respectively.

Panel A: 5-year spreads

	Credit default swap			Asset swap		
Sovereign	p_{ADF}	p_{PP}	KPSS stat.	p_{ADF}	p_{PP}	KPSS stat.
France	0.88	0.91	1.35	0.01	0.00	5.57
Germany	0.27	0.28	1.52	0.02	0.00	2.94
Greece	0.91	0.87	6.21	0.98	0.76	6.67
Ireland	0.48	0.48	2.87	0.13	0.12	7.04
Italy	0.45	0.62	2.24	0.01	0.00	3.73
Portugal	0.80	0.78	3.46	0.07	0.02	5.22
Spain	0.50	0.42	4.45	0.22	0.00	5.60

Panel B: 10-year spreads

	Credit default swap			Asset swap		
Sovereign	p_{ADF}	p_{PP}	KPSS stat.	p_{ADF}	p_{PP}	KPSS stat.
France	0.66	0.76	2.31	0.00	0.00	8.00
Germany	0.92	0.75	2.18	0.14	0.00	7.54
Greece	0.92	0.93	7.23	0.74	0.83	8.49
Ireland	0.47	0.28	4.68	0.20	0.62	9.58
Italy	0.31	0.30	3.36	0.06	0.23	6.77
Portugal	0.72	0.66	4.26	0.05	0.07	7.92
Spain	0.68	0.37	5.83	0.01	0.02	9.21

Table C.2: Unit root and stationarity tests in levels - crisis

The table reports the statistics of unit root and stationarity tests for the period from April 2010 to end 2011. Further details are presented in Table C.1.

Panel A: 5-year spreads

	Credit default swap			Asset swap		
Sovereign	p_{ADF}	p_{PP}	KPSS stat.	p_{ADF}	p_{PP}	KPSS stat.
France	0.80	0.74	7.57	0.15	0.18	4.43
Germany	0.80	0.62	7.35	0.60	0.31	7.16
Greece	1.00	1.00	7.79	0.00	0.00	9.67
Ireland	0.30	0.29	10.12	0.06	0.19	9.01
Italy	0.77	0.71	7.17	0.67	0.35	8.45
Portugal	0.79	0.69	10.80	0.26	0.14	11.29
Spain	0.11	0.08	7.83	0.03	0.03	7.51

Panel B: 10-year spreads

	Credit default swap			Asset swap		
Sovereign	p_{ADF}	p_{PP}	KPSS stat.	p_{ADF}	p_{PP}	KPSS stat.
France	0.99	0.98	7.94	0.49	0.81	5.21
Germany	0.48	0.49	4.10	0.59	0.09	6.58
Greece	0.94	0.97	8.68	0.17	0.00	5.36
Ireland	0.10	0.26	10.44	0.01	0.02	9.30
Italy	0.92	0.90	6.65	0.82	0.46	8.46
Portugal	0.77	0.79	11.26	0.01	0.09	11.26
Spain	0.86	0.73	8.02	0.19	0.15	8.51

D Cointegration analysis

We test for a long-run relationship in the form of cointegration between bond and CDS market credit premia using the tests of Phillips and Ouliaris (1990) and Johansen (1988).

We view two series as cointegrated if either the null hypothesis of no cointegration is rejected using the Johansen or the Phillips-Ouliaris methodology. We use the Johansen test with intercept but no deterministic trend in the co-integrating equation. We use the Schwarz information criterion to estimate the optimal lag length for the Johansen test.

The test results indicate that in all cases, the CDS and the ASW spread series are cointegrated.

Table D.1: Cointegration - p-values, pre-crisis

This table reports the probabilities in decimals obtained from the Johansen cointegration and the Phillips-Ouliaris cointegration tests for the period from January 2008 to end-March 2010. For the Johansen test a constant is included in the co-integrating equation and the number of lags in the vector autoregression is optimized using the Schwarz information criterion. The Phillips-Ouliaris tests for no cointegration under the null hypothesis by estimating the long-term equilibrium relationship from a regression of CDS_t on ASW_t or from a regression of ASW_t on CDS_t among the levels of the time series. The column header ASW and CDS indicates which variable is used as dependent variable in the test.

Panel A: Johansen test

	Trace test				Maximum eigenvalue test			
	5-year		10-year		5-year		10-year	
	None	at most 1	None	at most 1	None	at most 1	None	at most 1
Sovereign								
France	0.000	0.435	0.003	0.612	0.000	0.435	0.001	0.612
Germany	0.143	1.000	0.159	0.664	0.039	1.000	0.104	0.664
Greece	0.001	0.786	0.000	0.441	0.000	0.786	0.000	0.441
Ireland	0.022	0.949	0.015	0.557	0.005	0.949	0.008	0.557
Italy	0.004	0.517	0.001	0.944	0.002	0.517	0.000	0.944
Portugal	0.001	0.354	0.000	0.728	0.001	0.354	0.000	0.728
Spain	0.024	0.783	0.000	0.618	0.009	0.783	0.000	0.618

Panel B: Phillip-Ouliaris test

	τ -statistics				z-statistics			
	5-year		10-year		5-year		10-year	
	CDS	ASW	CDS	ASW	CDS	ASW	CDS	ASW
Sovereign								
France	0.182	0.002	0.935	0.000	0.379	0.026	0.935	0.000
Germany	0.023	0.001	0.393	0.053	0.147	0.043	0.511	0.138
Greece	0.006	0.024	0.001	0.001	0.002	0.004	0.002	0.002
Ireland	0.002	0.001	0.017	0.016	0.028	0.022	0.080	0.076
Italy	0.000	0.000	0.000	0.000	0.005	0.000	0.001	0.000
Portugal	0.001	0.000	0.039	0.004	0.022	0.008	0.035	0.008
Spain	0.523	0.000	0.010	0.000	0.585	0.021	0.028	0.002

Table D.2: Cointegration - p-values, crisis

This table reports the probabilities in decimals obtained from the Johansen cointegration and the Phillips-Ouliaris cointegration tests for the period from April 2010 to end 2011. Further details are presented in Table D.1.

Panel A: Johansen test

	Trace test				Maximum eigenvalue test			
	5-year		10-year		5-year		10-year	
Sovereign	None	at most 1	None	at most 1	None	at most 1	None	at most 1
France	0.149	0.732	0.022	0.057	0.086	0.732	0.097	0.057
Germany	0.001	0.682	0.975	0.955	0.000	0.682	0.940	0.955
Greece	0.984	0.978	0.016	0.990	0.951	0.978	0.003	0.990
Ireland	0.011	0.104	0.050	0.224	0.030	0.104	0.077	0.224
Italy	0.209	0.721	0.168	0.516	0.134	0.721	0.145	0.516
Portugal	0.000	0.312	0.360	0.374	0.000	0.312	0.458	0.374
Spain	0.023	0.130	0.326	0.441	0.054	0.130	0.364	0.441

Panel B: Phillip-Ouliaris test

	τ -statistics				z-statistics			
	5-year		10-year		5-year		10-year	
Sovereign	CDS	ASW	CDS	ASW	CDS	ASW	CDS	ASW
France	0.951	0.106	0.008	0.401	0.945	0.103	0.163	0.319
Germany	0.000	0.000	0.028	0.015	0.001	0.001	0.102	0.074
Greece	0.000	0.000	0.019	0.034	0.000	0.000	0.012	0.019
Ireland	0.034	0.024	0.000	0.000	0.008	0.006	0.002	0.003
Italy	0.002	0.001	0.000	0.000	0.014	0.011	0.000	0.000
Portugal	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Spain	0.000	0.000	0.000	0.000	0.001	0.000	0.000	0.000

E TVECM coefficients

Table E.1: TVECM coefficients

This table reports the TVECM coefficient β_0 for the 5-year and 10-year tenor. The coefficient β_1 was set to unity in our analysis. The pre-crisis period lasts from January 2008 to end-March 2010. The crisis period starts in April 2010 and ends end-December 2011. Numbers are only reported when a significant threshold was found and the error correction term is significant at 95% confidence level. For the crisis period in Germany no threshold model have been found for the German data (superscript ⁺).

	5-year tenor		10-year tenor	
Sovereign	pre-crisis	crisis	pre-crisis	crisis
France	148.87	169.29	64.87	97.55
Germany ⁺	9.00	-	-61.91	32.69
Greece	45.84	306.71	26.14	5.00
Ireland	25.83	226.14	-22.15	47.33
Italy	49.39	81.56	-5.74	59.51
Portugal	35.58	86.51	20.69	-10.00
Spain	87.83	106.38	40.92	92.55

F Basis volatility

Table F.1: Volatility basis - 5-year tenor

The table shows the volatility based on log changes and in bps. The pre-crisis period starts in January 2008 and ends in March 2010. The crisis period begins in April 2010 and our data ends in December 2011.

	pre-crisis		crisis	
Sovereign	lower regime	upper regime	lower regime	upper regime
France	2.14	1.25	1.08	1.29
Germany	1.01	1.08		
Greece	26.71	4.81	11.11	9.69
Ireland	2.40	2.31	7.27	3.56
Italy	2.18	1.72	2.26	1.99
Portugal	2.86	2.43	5.12	5.01
Spain	1.73	1.67	2.44	1.65

Table F.2: Volatility basis - 10-year tenor

The table shows the volatility based on log changes and in bps. The pre-crisis period starts in January 2008 and ends in March 2010. The crisis period begins in April 2010 and our data ends in December 2011.

	pre-crisis		crisis	
Sovereign	lower regime	upper regime	lower regime	upper regime
France	3.03	2.07	1.21	1.33
Germany	1.43	1.30	1.23	0.91
Greece	3.85	3.22	4.05	2.62
Ireland	2.84	2.62	4.24	2.21
Italy	3.05	2.15	2.31	2.23
Portugal	3.05	2.36	3.69	3.84
Spain	2.21	1.49	2.48	2.05